Smart Drip Irrigation for Alfalfa

A low-cost Arduino + IoT controller that senses field conditions, decides when to irrigate, and proves actuation through relay, LCD, and telemetry.

Sense | Decide | Actuate | Display / log

Project presentation

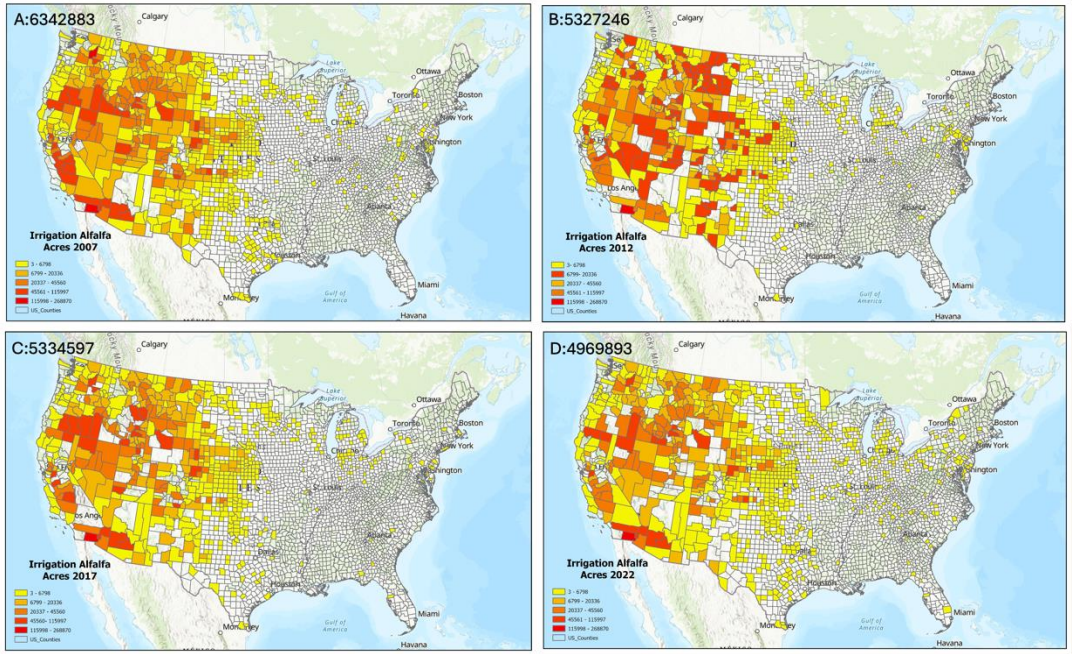
Why this project matters

Drought pressure, irrigation dependence, and a practical low-cost countermeasure

Field context

- Alfalfa is a major U.S. forage crop.
- Western systems face drought, limited water supply, and high evaporative demand.
- That makes irrigation timing, pump protection, and field visibility especially important.

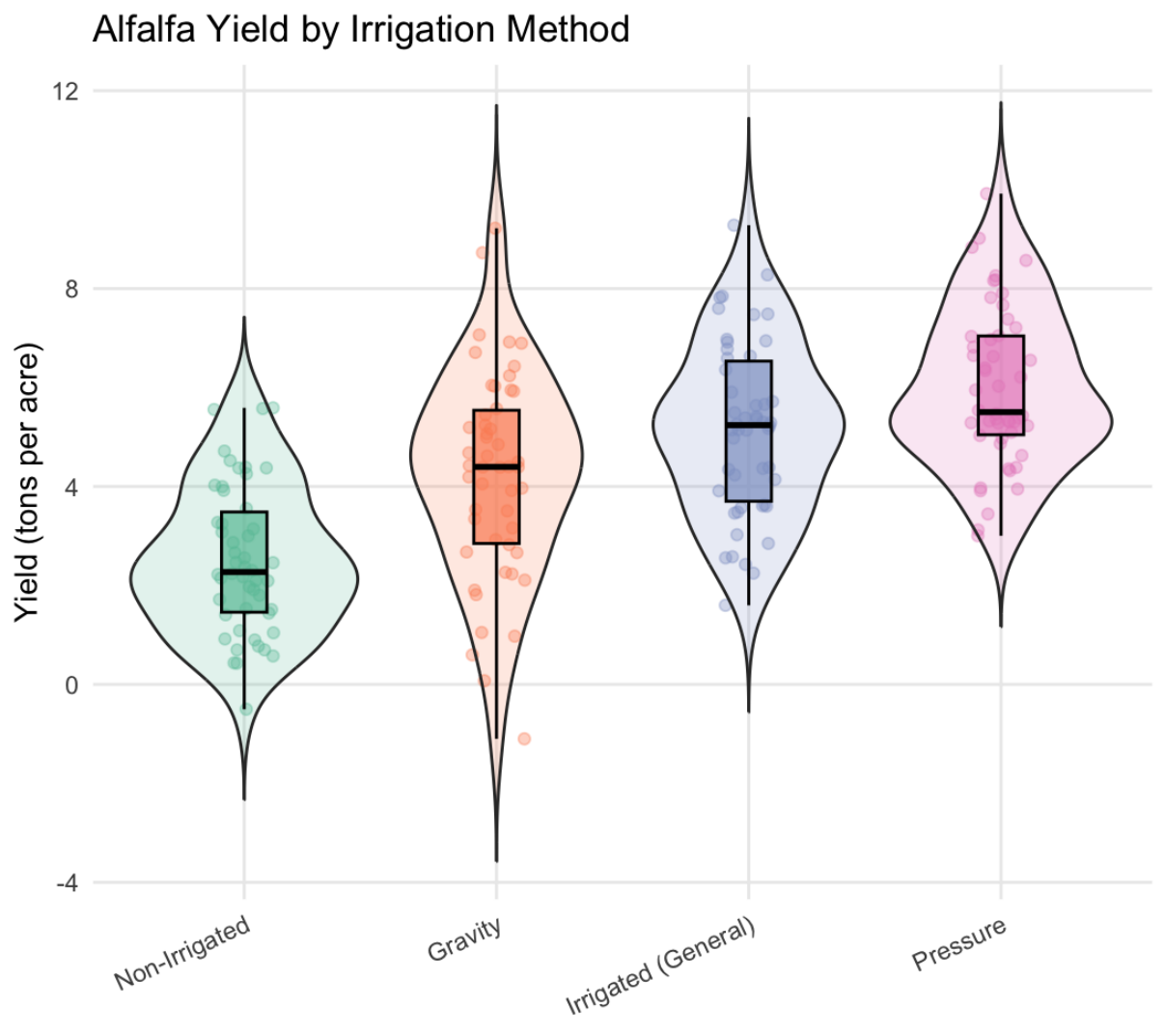
Control opportunity
Use low-cost sensing + control to trigger irrigation earlier when fields turn hot and dry.

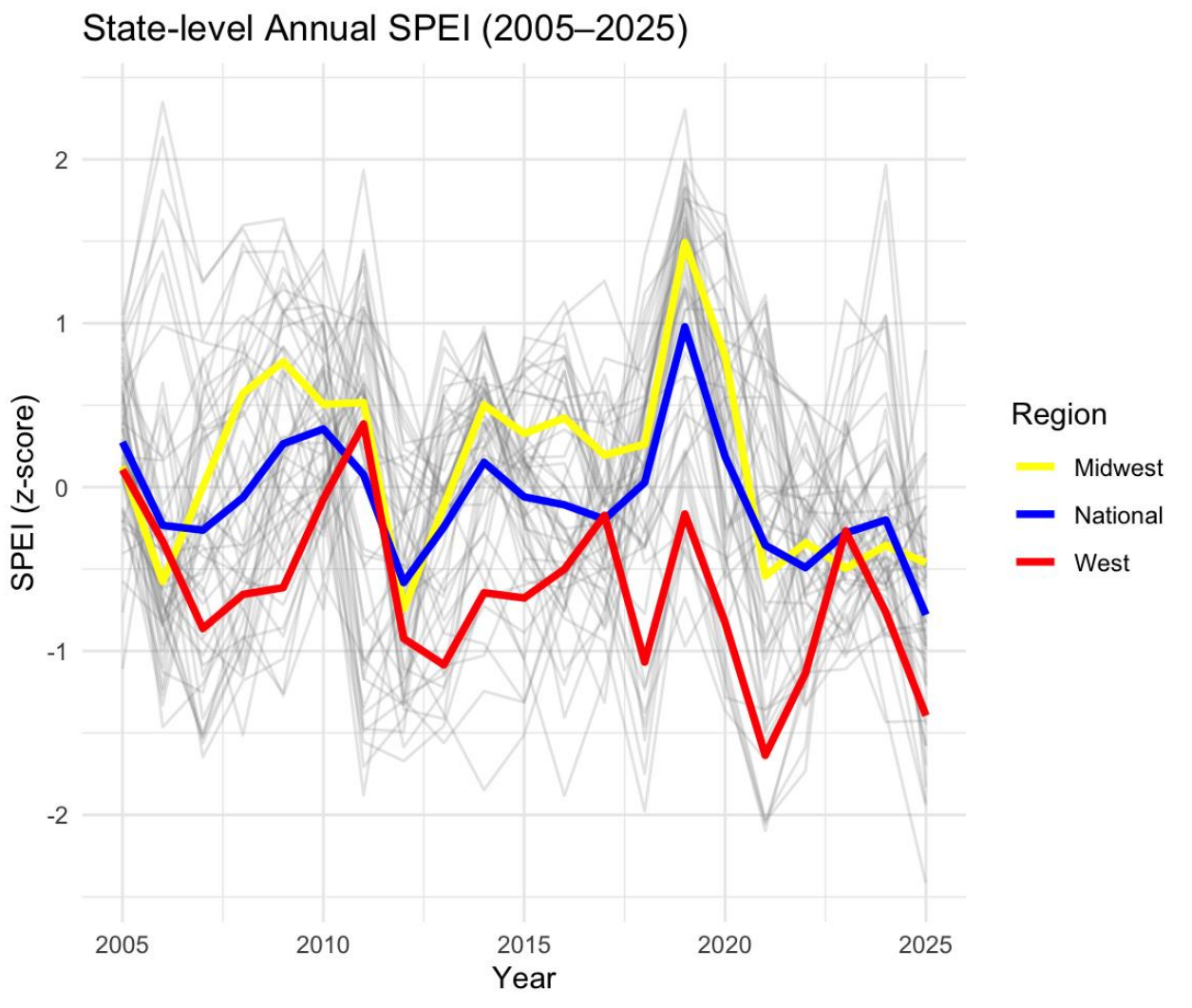


The project page pairs the irrigation prototype with western alfalfa / dairy geography, reinforcing why irrigation control matters most in water-stressed production corridors.

Prototype scope
Arduino Mega2560 + soil moisture sensor + DHT + LCD + relay + ESP-32S telemetry.

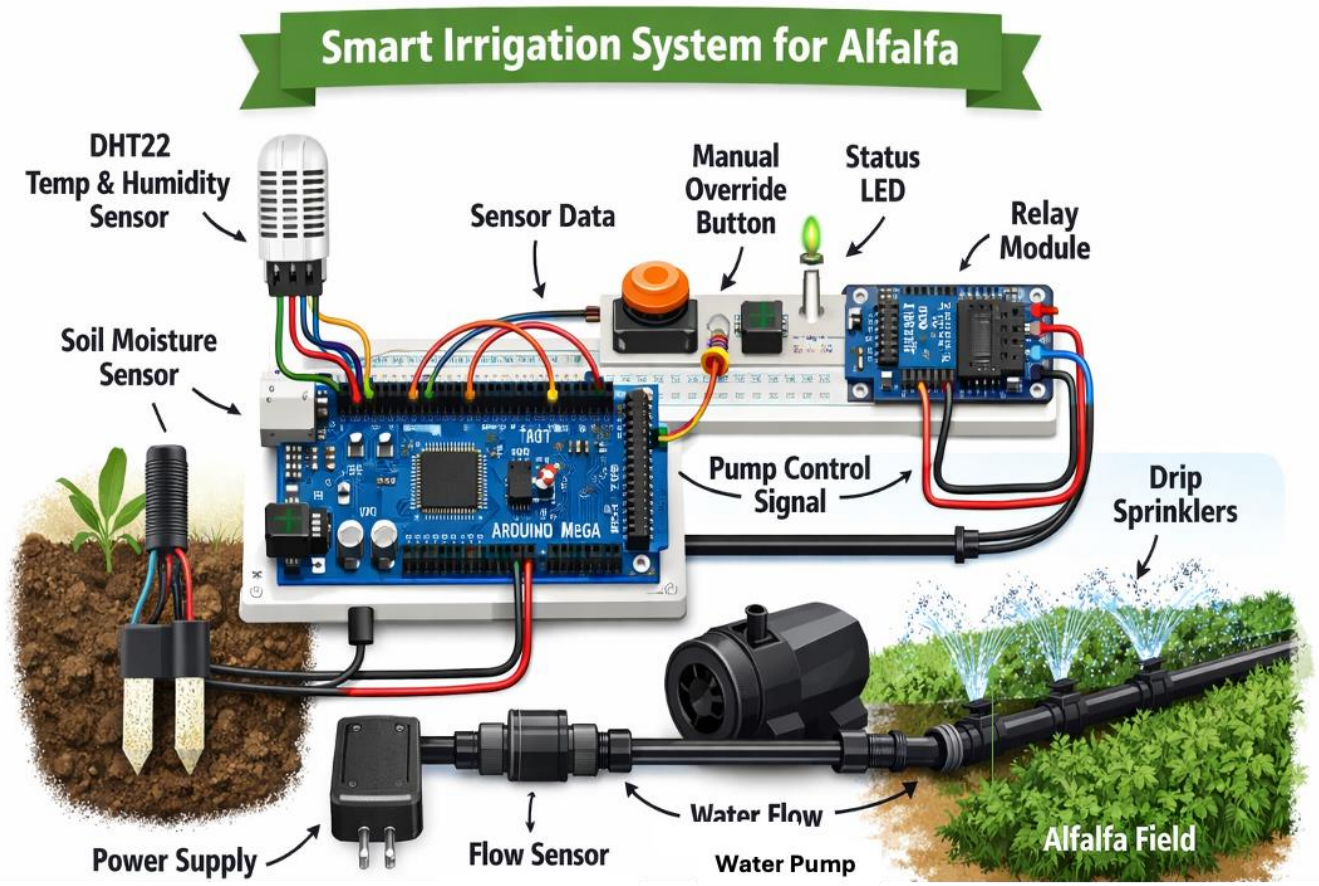
Why this project matters





Prototype architecture

Low-cost hardware with a clear control layer, sensing layer, and actuation path



Core stack

Mega2560

Soil sensor

Relay

DHT

Sensors: soil moisture (A0), water level (A1), DHT temperature/humidity (D2)

Control:
Mega2560 remains the reliable local controller

Actuation / UI:
relay on D8, LCD status display, keypad/manual button

Connectivity:
ESP-32S serial link for future dashboard / phone integration

Low-cost build | Modular I/O | IoT-ready

Control logic and safeguards

The controller uses hysteresis, environmental biasing, and fail-safe limits

1 Sense

Read soil ADC, water level, temperature, and humidity on a repeating interval.

2 Decide

Compute LOW / HIGH thresholds and shift them earlier when it is hot and dry.

3 Actuate

Pump turns ON below LOW and OFF above HIGH - a classic hysteresis loop.

4 Protect

Low water level, sensor faults, max runtime, and minimum off-time all force safe behavior.

Key thresholds

- LOW 30%
- HIGH 40%
- hot >= 30 C
- RH <= 30%
- bias +3
- MAX_ON 10 min

Implementation snippet

```
LOW=30; HIGH=40;
if (hot && dry) { LOW += 3; HIGH += 3; }
if (!pumpOn && soilPct < LOW) pumpOn();
if ( pumpOn && soilPct > HIGH) pumpOff();
```

Hard stops also force pump OFF on low water, bad readings, max runtime, or minimum off-time violations.

Operator interface and telemetry

Local controls, LCD feedback, and serial commands make the prototype testable

Local controls

- LCD shows startup, pump state, soil %, water level, and mode.
- Manual controls start / stop irrigation and switch AUTO or MANUAL.
- Numeric entry + # lets the operator set delivered volume.

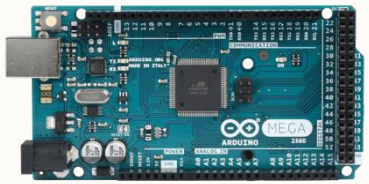
```
IRR:ON   S:28%
WL:520   A
```

Remote monitoring

The ESP32 serial link makes the controller observable and extensible for dashboards, weather feeds, and phone commands.

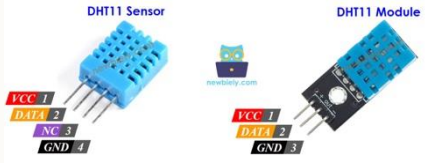
AUTO | MAN | STOP | VOL=... | START

```
mode=AUTO,soil_raw=702,soil_pct=29.6,wl_raw=520,wl_pct=72.3,temp_c=31.1,humidity=27.2,pump=ON,vol_ml=100,manual=0
```



Mega serial monitor

Relay actuation



Sensor + Wi-Fi extension

Roadmap and project value

What the prototype already proves - and how it can evolve into a fuller SDI system

Already demonstrated

- End-to-end loop from sensing to relay actuation
- Operator-readable LCD status and manual override
- Telemetry stream for monitoring and debugging
- A practical proof-of-concept using accessible hardware

Next build steps

- Add SD-card logging and RTC time stamps
- Implement “no watering at midday” scheduling rule
- Connect Wi-Fi dashboard / phone control through ESP32
- Pull weather data for more adaptive irrigation decisions

Why it matters

For irrigation-intensive alfalfa systems, this kind of controller is a practical base for tighter water timing, better observability, and future decision support.

Water-aware | Debuggable | Scalable

Thanks
Welcome comments and questions

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